

SUMMARY

Researcher with expertise in Generative AI, 3D/4D Reconstruction, Self-Supervised Learning, and Multi-Modal Foundational Models, with experience in both academic and industry settings.

EDUCATION

- **The University of Edinburgh** Edinburgh, UK
PhD in Computer Science *Sep 2021 – June 2025 (Thesis Submitted)*
- **Technical University of Munich** Munich, Germany
MSc in Computer Science *Oct 2018 – Dec 2020*
- **Istanbul Technical University** Istanbul, Turkey
BSc in Computer Science *Sep 2013 – Jun 2017*

EXPERIENCE

- **Meta AI** California, USA
Research Scientist Intern *Aug 2023 – Feb 2024*
 - **3D and Generative Learning:** Developed methods to integrate self-supervised learning and 3D generative modeling, focusing on mutual benefits for reconstruction and synthesis tasks.
- **Institute for Adaptive and Neural Computation** Edinburgh, UK
Postgraduate Researcher *Sep 2021 – Present*
 - **Shape and Semantics in ML:** Developed models for articulated 3D object reconstruction and explored the role of shape in semantic tasks, including fine-grained classification and correspondence.
- **TUM Computer Vision Lab** Munich, Germany
Research Assistant *Oct 2018 – Jun 2021*
 - **Scene Understanding:** Worked on Lidar-based semantic segmentation, instance segmentation, and multi-object tracking.
- **Coriolis Pharma** Munich, Germany
Student Machine Learning Engineer *Sep 2019 – Apr 2020*
 - **Applied Machine Learning:** Developed various deep learning solutions for flow imaging microscopy data.
- **Bunsar** Istanbul, Turkey
Machine Learning Engineer *Feb 2017 – Sep 2018*
 - **Applied Computer Vision:** Developed various computer vision solutions, including image retrieval, object detection/tracking, and visual recommendation systems for fashion retailers.

SELECTED PUBLICATIONS

- D. Duolikun, **M. Aygun**, C. Li, H. Bilen, and O. Mac Aodha. “DepthCues: Evaluating Monocular Depth Perception in Large Vision Models.” **CVPR 25’**
- **M. Aygun**, O. Mac Aodha. “SAOR: Single-view Articulated Object Reconstruction.” **CVPR 2024**
- **M. Aygun**, O. Mac Aodha. “Demystifying Unsupervised Semantic Correspondence Estimation.” **ECCV 2022**
- **M. Aygun** et al. “4D Panoptic Lidar Segmentation.” **CVPR 2021**
- **M. Aygun**, Y. Aytar, H. K. Ekenel. “Exploiting Convolution Filter Patterns for Transfer Learning.” **ICCV 2017 Workshop (Honourable Mention Award)**

SKILLS

- Python, C/C++, PyTorch (3D), OpenCV, NumPy, scikit-learn, Matlab, Linux, Git, AWS, Kubernetes, Docker